Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2026-27 (Odd)**	**Course Name**	**LAB: Optimization technique**
Semester	**7th**	**Course Code**	**23ADS1708**
Roll No	**39**	**Name of Student**	**Hrutik Gujarkar**

Practical Number	**8**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To determine the shortest paths using the Floyd–Warshall Algorithm.
Problem Definition	To find the shortest distance between every pair of vertices in a weighted directed graph with 5 vertices using the Floyd–Warshall Algorithm. The edges (with weights) are: 1→2 (3), 1→3 (8), 1→5 (7), 2→4 (1), 3→2 (4), 4→1 (2), 4→3 (5), 5→4 (6).
Theory (100 words)	The Floyd–Warshall algorithm is a dynamic programming method that finds the shortest paths between all pairs of vertices in a weighted graph. It works with negative edge weights but not with negative cycles. Let d[i][j] be the shortest known distance from vertex i to vertex j. The algorithm takes each vertex k in turn as a possible intermediate vertex and updates d[i][j] = min(d[i][j], d[i][k] + d[k][j]). After all n vertices have been considered, d holds the shortest distances. The time complexity is $O(n^3)$ and the space complexity is $O(n^2)$. A negative value on the diagonal indicates a negative cycle.

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	Step-by-Step Method 1. Represent the graph as an n × n matrix: 0 on the diagonal, the edge weight where an edge exists, and ∞ where there is no edge. 2. Initialize the distance matrix D with this matrix. 3. For k = 1 to n, take vertex k as the intermediate vertex. 4. For every pair (i, j), update d[i][j] = min(d[i][j], d[i][k] + d[k][j]). 5. After all n iterations, the matrix contains the shortest distance between every pair of vertices. 6. Check the diagonal of the matrix for negative values to detect a negative cycle. 7. Reconstruct the actual shortest paths using the next-vertex matrix.
Procedure and Execution (100 Words)	Algorithm: 1. Define the weighted graph as an adjacency matrix. 2. Initialize the distance matrix and the next-vertex matrix. 3. For each intermediate vertex k, for each pair (i, j): 4. if d[i][k] + d[k][j] < d[i][j], update d[i][j] and next[i][j]. 5. Check the diagonal for a negative cycle. 6. Display the final shortest distance matrix. 7. Reconstruct and display the shortest path for every pair of vertices.

	Code:
	`INF = float("inf")`
	
	`# Weighted directed graph (adjacency matrix), vertices 1 to 5`
	`graph = [`
	`  [0,  3,  8, INF,  7],`
	`  [INF, 0, INF,  1, INF],`
	`  [INF, 4,  0, INF, INF],`
	`  [2, INF,  5,  0, INF],`
	`  [INF, INF, INF,  6,  0],`
	`]`
	`n = len(graph)`
	
	`def show(matrix, title):`
	`  print(title)`
	`  print("  " + "".join(f"{j+1:>5}" for j in range(n)))`
	`  for i in range(n):`
	`  row = ""`
	`  for v in matrix[i]:`
	`  row += f"{('INF' if v == INF else int(v)):>5}"`

```
        print(f"{i+1:>4}  {row}")
    print()

# Distance matrix and next-vertex matrix (for path reconstruction)
dist = [row[:] for row in graph]
nxt = [[j if graph[i][j] != INF and i != j else None
        for j in range(n)] for i in range(n)]

show(dist, "Initial Distance Matrix:")

# Floyd-Warshall algorithm (k = intermediate vertex)
for k in range(n):
    for i in range(n):
        for j in range(n):
            if dist[i][k] + dist[k][j] < dist[i][j]:
                dist[i][j] = dist[i][k] + dist[k][j]
                nxt[i][j] = nxt[i][k]

show(dist, "Final Shortest Distance Matrix:")

# Negative cycle check
if any(dist[i][i] < 0 for i in range(n)):
    print("Graph contains a negative cycle")

# Path reconstruction
def get_path(u, v):
    if nxt[u][v] is None:
        return None
    path = [u + 1]
    while u != v:
        u = nxt[u][v]
        path.append(u + 1)
    return path

print("Shortest Paths:")
for i in range(n):
    for j in range(n):
        if i != j:
            p = get_path(i, j)
            if p is None:
                print(f"{i+1} -> {j+1}: no path")
            else:
                route = " -> ".join(map(str, p))
                print(f"{i+1} -> {j+1}: cost = {int(dist[i][j])}, "
                      f"path = {route}")
```

Output:

```
Initial Distance Matrix:
          1    2    3    4    5
   1      0    3    8  INF    7
   2    INF    0  INF    1  INF
   3    INF    4    0  INF  INF
   4      2  INF    5    0  INF
   5    INF  INF  INF    6    0

Final Shortest Distance Matrix:
          1    2    3    4    5
```

	1 0 3 8 4 7 2 3 0 6 1 10 3 7 4 0 5 14 4 2 5 5 0 9 5 8 11 11 6 0 Shortest Paths: 1 -> 2: cost = 3, path = 1 -> 2 1 -> 3: cost = 8, path = 1 -> 3 1 -> 4: cost = 4, path = 1 -> 2 -> 4 1 -> 5: cost = 7, path = 1 -> 5 2 -> 1: cost = 3, path = 2 -> 4 -> 1 2 -> 3: cost = 6, path = 2 -> 4 -> 3 2 -> 4: cost = 1, path = 2 -> 4 2 -> 5: cost = 10, path = 2 -> 4 -> 1 -> 5 3 -> 1: cost = 7, path = 3 -> 2 -> 4 -> 1 3 -> 2: cost = 4, path = 3 -> 2 3 -> 4: cost = 5, path = 3 -> 2 -> 4 3 -> 5: cost = 14, path = 3 -> 2 -> 4 -> 1 -> 5 4 -> 1: cost = 2, path = 4 -> 1 4 -> 2: cost = 5, path = 4 -> 1 -> 2 4 -> 3: cost = 5, path = 4 -> 3 4 -> 5: cost = 9, path = 4 -> 1 -> 5 5 -> 1: cost = 8, path = 5 -> 4 -> 1 5 -> 2: cost = 11, path = 5 -> 4 -> 1 -> 2 5 -> 3: cost = 11, path = 5 -> 4 -> 3 5 -> 4: cost = 6, path = 5 -> 4
Output Analysis	The program computed the shortest distance between all pairs of vertices. Distances that were initially infinite became finite through intermediate vertices. For example, d(1, 4) = 4 through 1 → 2 → 4 (3 + 1), d(5, 2) = 11 through 5 → 4 → 1 → 2 (6 + 2 + 3), and d(3, 5) = 14 through 3 → 2 → 4 → 1 → 5 (4 + 1 + 2 + 7). All diagonal values are 0, so the graph has no negative cycle.
Link of student Github profile where lab assignment has been uploaded	https://github.com/HrutikGujarkar/Optimizaton-techniques-Practicals
Conclusion	The shortest paths between all pairs of vertices were successfully determined using the **Floyd–Warshall Algorithm**. The final distance matrix and the corresponding shortest routes were obtained for the given 5-vertex graph.

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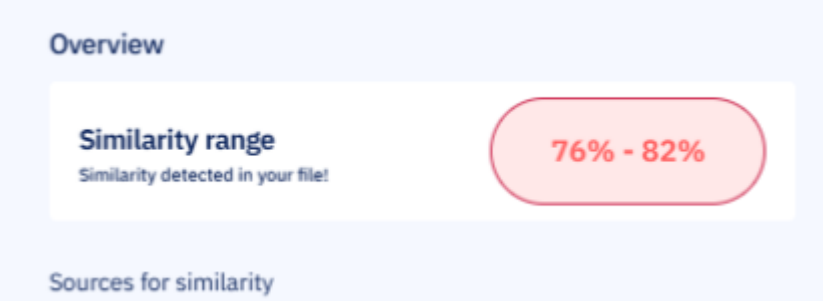
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